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Optimizing Nitrogen Fertilizer Application in Paddies Using Machine Learning for Leaf Colour Detection

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We certify that this proposal/thesis does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any university, and to the best of our knowledge and belief, it does not contain any material previously published or written by another person except where due reference is made in the text.

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List of Abbreviations

- **AI:** Artificial Intelligence
- **CLAHE:** Contrast Limited Adaptive Histogram Equalization
- **CNN:** Convolutional Neural Network
- **DFD:** Data Flow Diagram
- **HDR:** High Dynamic Range
- **HSV:** Hue, Saturation, Value (color space)
- **IRRI:** International Rice Research Institute
- **JPEG:** Joint Photographic Experts Group (image format)
- **LAB:** Lightness, A, B (color space)
- **LCC:** Leaf Color Chart
- **ML:** Machine Learning
- **N:** Nitrogen
- **RGB:** Red, Green, Blue (color model)
- **SPAD:** Soil Plant Analysis Development (chlorophyll meter)
- **SVM:** Support Vector Machine
- **UML:** Unified Modeling Language

1.Abstract

Nitrogen is one of the principal nutrients for rice farming and is involved in many functions like the synthesis of chlorophyll, vegetative growth, photosynthesis, and grain production. Yet, mismanagement of nitrogen fertilizer either in the form of underapplication or overapplication has far-reaching consequences on agriculture and the environment. Nitrogen deficiency causes weak plants and reduced yields, while overapplication causes water pollution, greenhouse gas emission, and high production costs.

Farmers have traditionally depended on tools like the Leaf Color Chart (LCC) to approximate nitrogen content, but they are susceptible to human error, inconsistency, and moderate accuracy. While more precise solutions are available, like SPAD meters, their expense and technological requirements render them unsuitable for smallholder farmers, particularly in developing countries. This research presents a mobile application based on machine learning that offers an affordable, precise, and real-time nitrogen deficiency detection technique in rice leaves. Essentially, the application employs a pre-optimized DenseNet121 Convolutional Neural Network (CNN) model trained using thousands of images of rice leaves with annotations. The model categorizes images of leaves into four different nitrogen status categories: severe, moderate, and mild deficiency, and sufficient nitrogen status. It has a 98% high classification accuracy even when used to classify images taken from other phone models and with varying lighting conditions.

For model robustness and image quality improvement, the preprocessing pipeline is also equipped with Contrast Limited Adaptive Histogram Equalization (CLAHE) for delivering stable performance under varying field conditions. The mobile application is built on the Flutter platform and is cross-platform, low-end Android device compatible, and lightweight in design. It delivers results within two seconds and gives Specific fertilizer recommendations. With its integration of artificial intelligence and mobile technology, the study provides a scalable, low-cost, and sustainable precision nitrogen management solution with particular benefits for smallholder rice farmers in resource-constrained environments.

Keywords: Nitrogen management, rice farming, machine learning, convolutional neural network, mobile app, precision agriculture, sustainable agriculture.

2. Introduction

2.1 Overview of the Proposed Study

Nitrogen is an essential rice crop nutrient that has direct effects on photosynthesis, growth, and grain yield. However, inefficient management of nitrogen is still a main issue in paddy cultivation, resulting either in yield reduction (due to deficiency) or environmental pollution (due to over-use). Conventional nitrogen monitoring techniques, like the Leaf Color Chart (LCC), are based on subjective visual comparison, which is prone to human error and inconsistency. Meanwhile, precision equipment like SPAD chlorophyll meters are much too expensive for smallholder producers, which are the majority of rice farmers in developing countries.

This study overcomes these limitations by proposing a machine learning-based mobile application that has the capability to automatically detect nitrogen deficiency in rice leaves. The application categorizes leaf color in relation to nitrogen status as severe deficiency, moderate deficiency, mild deficiency, and sufficiency using Convolutional Neural Networks (CNNs) based on standard LCC guidelines.

To facilitate robustness, the model uses state-of-the-art image preprocessing methods for normalizing variations due to various smartphone cameras and light conditions. The application then gives location-specific fertilizer suggestions in real time to enable farmers to make efficient nitrogen applications. By combining precision agriculture with widely accessible mobile technology, the project aims to democratize nitrogen management, making it affordable, precise, and scalable for small-scale farmers in Sri Lanka and elsewhere in other rice-growing regions.

2.2 Project Scope

Scope of the project has some clearly defined parameters. The research is exclusively on rice cultivation with particular emphasis on widely cultivated varieties in Sri Lanka. The intended user group is small-scale farmers who generally cultivate less than two hectares of land and are not equipped with sophisticated agricultural technologies. The system takes input in the form of images of leaves captured using regular smartphone cameras, with processing aimed to be executed on both high-end and low-cost mobile phones.

Outputs include a nitrogen deficiency class and precise fertilizer application suggestions depending on local conditions. The suggestions are according to plant growth stage and dominant soil types in the locality. The project presently limits its scope to nitrogen management and does not cover other nutrient deficiencies, although the design allows for easy future expansion to other nutrients. Among the design specifications is compatibility with low-end smartphones in order to reach the intended user group.

2.3 Motivation for Study

The inspiration for this study comes from three salient problems that cut across agricultural productivity, environmental sustainability, and economic fairness. Food security worldwide is an acute concern, and rice is the main staple for roughly half of the global population. Future population growth will require colossal improvements in rice yield, which must be made without parallel expansions in the area planted. Better nitrogen management provides one extremely promising avenue to improve yield and optimize the utilization of available inputs.

Environmental factors prove to be just as powerful a motivator. Inefficient use of fertilizers is one of the major causes of water pollution via nitrate leaching and global warming via nitrous oxide emissions. The environmental effect does not stay local to agriculture, with downstream ecosystems and groundwater resources being impacted. On the economic front, the price of precision ag equipment gives systemic disadvantage to smallholder farmers, who make up the majority of rice producers in the developing world. This technology imbalance strengthens yield differentials in limiting rural households' income prospects.

2.4 Rationale and Justification

The rationale for this study arises from noted loopholes in current nitrogen assessment methods. Conventional LCC methods, though inexpensive, exhibit unreliable outcomes due to their dependency on subjective human opinion. Field tests have recorded over 40% difference in fertilizer suggestion for the same leaf samples through manual color matching methods. On the other hand, though SPAD meters give unbiased readings, their high cost eliminates about 85% of prospective users in developing agricultural systems.

Emerging digital solutions try to fill this gap; however, they are beset with inherent limitations. Existing mobile apps taken up for leaf color analysis tend to be poorly performing under different field conditions, with their accuracy being highly reliant on particular device characteristics and surrounding light. None of the existing solutions incorporate comprehensive recommendation mechanisms, offering mostly simple color analysis with no helpful guidance. In this paper, we strive to transcend these limitations by proposing a system that is explicitly tailored to normalize technical disparities and offer full decision-support functionality.

2.5 Research Problem

The underlying research issue is focused on three interlinked difficulties in rice nitrogen management. Firstly, subjectivity in manual evaluation techniques results in inconsistent fertilizer application, thus causing either under-application with yield loss or economically and ecologically expensive over-application. Secondly, the expense of precision measuring devices causes systemic obstacles to adoption for small-scale growers. Third, existing digital methods fail to adequately reflect the technical variability that occurs in real-world field conditions, particularly with respect to device heterogeneity and ambient illumination.

These difficulties are most pronounced in the developing world where rice farming is the cornerstone of food security and rural economies. Smallholder farmers in the developing world generally have limited access to agricultural extension services and are under tight capital availability constraints. The implications of poor nitrogen management therefore reach beyond the level of the individual farm, affecting regional food potential for production and environmental quality.

2.6 Research Objectives

The research aims for one overall objective supported by three technical objectives. The overall objective involves the development of an accurate, handy, and affordable nitrogen management system for smallholder rice farmers. The overall objective decomposes into technical specific targets beginning with the construction of a high-precision CNN model that can classify leaf images into four nitrogen status categories at over 95% accuracy.

The second specific objective involves creating an intelligent recommendation system that interprets classification outcomes into usable fertilizer recommendations. The system takes into account certain contextual parameters such as local soil type and crop growth stage in order to provide location-specific prescriptions. The third objective involves making these functionalities accessible via an intuitive mobile app that is performance-optimized for low-end smartphones common in the target user groups.

2.7 Hypotheses

H₁ : "A CNN model with CLAHE normalization will achieve >95% accuracy, outperforming manual LCC methods."

This hypothesis predicts the suggested Convolutional Neural Network (CNN) model, enhanced with Contrast Limited Adaptive Histogram Equalization (CLAHE) image normalization, to classify rice leaf nitrogen deficiency levels with an accuracy of more than 95%. This accuracy is predicted to be better than conventional manual Leaf Color Chart (LCC) methods, which continue to be vulnerable to human subjectivity and variability.

H₂ : "The mobile app will reduce fertilizer overuse by 15% in field trials compared to traditional practices."

This hypothesis reads that farmers using the mobile app will apply 15% less surplus nitrogen fertilizer compared to farmers using conventional LCC-based methods. This reduction is due to more precise deficiency detection and data-driven suggestions.

2.8 Research Questions

1. RQ1: How does the CNN model's accuracy compare to SPAD meters and manual LCCs?
2. RQ2: Which image preprocessing techniques best compensate for lighting and device variations?
3. RQ3: Can the app improve nitrogen-use efficiency in real-world farming conditions?

2.9 Significance of the Research

This work is important on several levels. Theoretically, it adds to the body of literature on uses of machine learning in agriculture in real-world settings, and specifically on tackling real-world problems of variability. On a practical level, the project illustrates the implementation of high-level technology into situations of resource paucity, and as such could act as a role model for other agricultural uses.

The social benefits are no less important as the technology can help bridge the precision agriculture divide between smallholder and large-scale commercial farmers. Environmentally, widespread adoption could translate to reduced fertilizer runoff and associated pollution. Economically, the system can improve farm-level profitability by increasing efficiency of input utilization while providing affordable accessibility through smartphone-based dissemination.

2.10 Expected Outcomes

Expected results are functional and technical. The CNN model is anticipated to exhibit classification accuracy of around 98% in controlled validation tests. The mobile app is built to run in times of less than two seconds on low-cost smartphones, with usability tests targeting 90% successful use without technical support. Field tests are planned to confirm at least a 15% increase in nitrogen-use efficiency compared to conventional practice, with corresponding savings in fertilizer costs and environmental footprint.

The system design overall will be documented in a way that future expansion to other types of crops and fertilizer classes is eased, with a basis for further generic application in precision agriculture. Not just the technical components, but also farmers' training materials will be deliverables so that adoption and correct usage can be enabled in target communities.

3. Literature Review

3.1 Theoretical Background

Since it affects grain production, vegetative development, and chlorophyll synthesis, nitrogen (N) is essential to rice agriculture. Its application is still a delicate balance, though, since both excessive and insufficient use can have detrimental ecological and agronomic effects. While nitrogen deficit can result in pale leaves and low production, excessive nitrogen leads to greenhouse gas emissions, groundwater pollution, and soil acidification (Zhang, Tao, X., J. & Chen, Y., 2020). Historically, techniques such as the Leaf Colour Chart (LCC) and SPAD meters have been used to regulate nitrogen in rice fields. Despite their worth, these approaches have drawbacks in terms of subjectivity, cost, and scalability, particularly for smallholder farmers in areas with limited resources (Sharma, A., Ghosh, R. & Das, S., 2021).

A new paradigm automated, picture-based assessment of leaf greenness utilising smartphone cameras and artificial intelligence has evolved as a result of advancements in mobile technology, machine learning, and image processing. This pattern is in line with the larger movement towards smart farming methods and precision agriculture (Sinha, 2024). Nowadays, a lot of systems use supervised machine learning or deep learning methods like convolutional neural networks (CNNs) to assess nitrogen status based on RGB and HSV colour values from leaf pictures (Siddique, M.A., Hasan, M.T. & Rahman, A., 2023).

3.2 Previous Work

3.2.1 Traditional Nitrogen Estimation Methods

The Kjeldahl method, which evaluates the total nitrogen content by acid digestion and distillation, is one of the chemical assays that started the evolution of traditional nitrogen estimate techniques in rice production decades ago. Despite its accuracy, this approach is still resource-intensive and unsuitable for field diagnostics in real time. Because they can measure chlorophyll quickly and without causing damage, portable instruments like SPAD meters have become popular as a way to get around these restrictions. Particularly, the SPAD-502 meter has demonstrated a strong relationship with rice nitrogen levels; under controlled settings, several investigations have reported R^2 values over 0.85 (Yuan, 2022). Unfortunately, smallholder farmers in low-income areas cannot afford SPAD meters, which can cost up to USD 300 (Chen, 2022). Furthermore, they are vulnerable to localised variations in chlorophyll content because of their tiny sensor area, which only records a small fraction of the leaf. The International Rice Research Institute (IRRI) developed the Leaf Colour Chart (LCC), which has proven to be an inexpensive substitute. In order to ascertain the proper rates of fertiliser application, farmers might use it to visually compare leaf greenness to a printed chart.

The LCC has poor reproducibility and subjectivity, despite its inexpensive cost. According to studies, the accuracy of assessments is greatly impacted by user judgement, ambient light levels, and chart deterioration with time. Nitrogen recommendations based on LCC readings differed by up to 25% across users in a comparative field trial by Aryal under various lighting conditions. The necessity for more impartial and uniform diagnostic instruments is highlighted by these discrepancies. To close this gap, smartphone-based imaging has emerged, providing digital nitrogen estimate via analysis of leaf images. Nevertheless, these techniques are still in their infancy and have comparable difficulties about image consistency and standardisation. The transition to intelligent, image-driven methods is thus accelerated by the limits of existing techniques, even if they have established the groundwork for in-field nitrogen assessment.

3.2.2 Leaf Colour as a Proxy for Nitrogen Status

Because nitrogen is necessary for the manufacture of chlorophyll, leaf colour, especially greenness, has been well proven to be a good indicator of rice's nitrogen content (Sinha, 2024). Nitrogen levels determined by SPAD or chemical tests have been found to strongly correlate with RGB colour values from digital photographs, according to recent research. For example, under controlled settings, (Wu, Y., Al-Jumaili, S.J., Al-Jumeily, D. and Bian, H., 2022) showed a correlation coefficient of 0.91 between the observed nitrogen content in rice and leaf greenness (as evaluated by the mean G value in RGB).

Since hue and chroma parameters provide better separation of green intensity and lessen sensitivity to brightness variations, a number of image-based nitrogen estimation techniques now rely on converting RGB values to other colour spaces, such as HSV or CIE Lab (Zhang, Tao, X., J. & Chen, Y., 2020). These techniques have proven successful in differentiating between plants that are nitrogen-sufficient and those that are nitrogen-deficient.

However, the shift from controlled to field settings has exposed significant drawbacks. Without preprocessing or calibration, nitrogen estimate becomes incorrect due to environmental factors such as shadows, leaf orientation, and natural lighting that produce inconsistent colour representation. The RGB and HSV values of even identical leaves taken under various lighting situations change greatly. According to some research, there might be a 20% variation in the average green intensity between photos shot in overcast, lunchtime, and morning sunlight.

Some researchers have included colour reference strips or white balance cards next to the leaf when taking pictures to lessen these effects. According to (Intaravanne, 2020), using a white reference card during field photography can minimise greenness variability by as much as 30%. Due to user non-compliance or a lack of training, this approach is frequently disregarded in practical applications, despite its efficacy (Paleari, 2019). Thus, even though leaf colour is still a potential signal for estimating nitrogen, a major research problem is still obtaining physiologically relevant and consistent colour data under a variety of situations. Image-based methods run the

danger of distorting biological reality in the absence of standardised imaging techniques or automated correction algorithms, which might result in inaccurate fertiliser recommendations.

3.2.3 Digital and Mobile-Based Nitrogen Detection Tools

The development of mobile applications for nitrogen quantification using leaf image processing has accelerated due to the growing availability of cell phones in rural and agricultural settings. These apps provide a low-cost, easily accessible substitute for conventional techniques, and they are especially helpful in areas with limited access to diagnostic instruments and extension services (Zhang, Tao, X., J. & Chen, Y., 2020). These include the IRRI-developed Nutrient Manager app, BaiKhao, and PocketN, which let users take pictures of rice leaves and get recommendations for nitrogen application based on colorimetric or machine learning analysis (Intaravanne, 2020).

These applications have been validated in experimental settings by several research. For example, under regulated illumination circumstances, BaiKhao demonstrated an accuracy of over 93% in nitrogen status categorisation (Intaravanne, 2020). Similarly, another measurement of PocketN in Pakistani rice fields revealed up to 88% concordance with nitrogen estimations based on SPAD. In practical applications, their accuracies drastically decrease. Reliability is decreased by noise introduced by variations in smartphone cameras, picture compression, and lighting conditions.

The absence of guided image capturing in the applications is yet another significant drawback. The majority of systems don't offer real-time photo quality feedback or user instructions. According to some research, their classification algorithms rejected about 30% of user-submitted photos because of misalignment or inadequate illumination. To improve user compliance and data consistency, it is crucial to incorporate guided capture methods, such as exposure alarms, framing help, or automatic quality checks.

Additionally, only a small percentage of apps use automated preprocessing to normalise collected pictures prior to analysis. Classification accuracy might be significantly increased by making minor adjustments such brightness modification, background removal, and white balance correction. According to early research, the inclusion of these preprocessing procedures improved the accuracy of nitrogen categorisation from 78% to 91% under a variety of illumination conditions. Ultimately, even though digital technologies are a huge advancement in nitrogen management, their efficacy depends on finding a solution to the fundamental problem of input unpredictability. For greater acceptance and effect in precision agriculture, these constraints must be addressed with improved user assistance, preprocessing pipelines, and adaptive machine learning models (Siddique, M.A., Hasan, M.T. & Rahman, A., 2023).

3.2.4 Machine Learning Applications for Leaf Colour Analysis

Since 2018, the use of machine learning (ML), especially deep learning, to analyse leaf colour for nitrogen estimation has been transformed. Because they can directly extract multiscale colour and texture information from raw picture pixels, Convolutional Neural Networks (CNNs) have

emerged as the preferred technique (Siddique, M.A., Hasan, M.T. & Rahman, A., 2023) and (Sharma, A., Ghosh, R. & Das, S., 2021). CNNs enable end-to-end learning from picture to classification by doing away with the requirement for manually created features.

PaddyNet, a CNN model trained on a sizable dataset of labelled rice leaf photos taken in semi-controlled settings, was first presented by (Siddique, M.A., Hasan, M.T. & Rahman, A.) in 2023. The model's accuracy in differentiating between leaves with and without nitrogen was above 94%. However, performance losses of more than 20% were seen when generalised to new datasets with various environmental circumstances or camera types. These drops were mostly caused by differences in colour tone, leaf placement, and shadow interference (Naik, 2020).

By using transfer learning techniques, which include fine-tuning pre-trained CNNs like ResNet50 or MobileNetV2 on datasets unique to rice, several studies tried to lessen this (Nugroho, 2022) and (Jaiswal, 2020). Although these models are more resilient to changes in the environment, they are nevertheless susceptible to distortions caused by cameras and harsh lighting. The use of hybrid models and ensemble learning has been another tactic. For example, improved classification stability by combining CNN-extracted features with support vector machines (SVM). To increase forecast accuracy, researchers have also begun adding meta-data to the machine learning process, such as the time of day, cloud cover, or smartphone model (Phogat, 2023).

One significant drawback still exists despite these developments: many models are trained on carefully chosen, well-lit photos, which makes them incompatible with actual deployment situations. Machine learning predictions run the danger of being influenced more by ambient artefacts than by actual physiological signals if picture inputs are not standardised by pre-capture instruction or post-capture normalisation.

3.2.5 Challenges of Image Standardisation for Colour Detection

One of the most urgent problems with leaf colour-based nitrogen assessment is still the absence of standardised procedures for picture collecting and preprocessing. Uncontrolled environmental factors such as background interference, shadowing, ambient light, and leaf surface reflectance continue to cause errors in the majority of image-based systems as of 2025 (Chen, 2022). Another degree of difficulty is introduced by smartphone variability, which includes anything from camera quality to built-in colour-enhancing algorithms (Wang, 2023).

Typical artefacts consist of:

- Colour temperature changes are caused by auto-white balance issues.
- High Dynamic Range (HDR) modifies saturation and contrast.
- Image quality is deteriorated by JPEG compression.

In green-channel sensitivity, which is essential to nitrogen measurement, these artefacts are especially harmful. Some researchers have suggested using physical calibration tools, such a

colour checker or grey card in the frame, to solve this (Fuentes, 2021) and (Gao, 2020). But in practical implementations, such conventions are still not always followed.

Software-based standardisation is used in other approaches. Despite their widespread use, histogram matching, gamma correction, and white balance normalisation frequently don't work across devices unless they are properly adjusted (Hasan, 2020). Leaf shadows or similar-coloured surrounds might confuse background segmentation, which is frequently required to separate the leaf. Although deep learning techniques for semantic segmentation, such as U-Net and Mask R-CNN, have shown promise in accuracy, they need a large amount of computation and annotated training data (Tang, 2021)

When applied to noisy field data, machine learning models trained on perfect photos may produce incorrect classifications if criteria are not maintained. For example, (Xu, 2023) discovered that 31% of the time, nitrogen-sufficient leaves captured in low light were incorrectly identified as deficient. On the other hand, leaves that naturally yellowed because of ageing instead of a nitrogen shortage were mistakenly marked for intervention.

As a result, picture standardisation is now a necessary condition for scalable and physiologically relevant nitrogen estimates rather than a side problem. The dependability of even the most sophisticated machine learning systems is compromised by its absence.

3.2.6 Advances in Image Preprocessing and Normalisation

The development of preprocessing pipelines to standardise leaf pictures for nitrogen estimate has advanced significantly between 2018 and 2025. These techniques seek to improve the reliability of physiological signals while reducing non-biological variation.

Important preprocessing methods include of:

- Brightness and contrast are normalised by histogram equalisation (Zhang, Tao, X., J. & Chen, Y., 2020).
- According to (Hasan, 2020) white balance adjustment is used to account for variations in illumination.
- switching to other colour spaces (Lab, HSV) to improve the separation of green channels (Sharma, A., Ghosh, R. & Das, S., 2021).
- techniques for leaf segmentation that use deep learning or edge detection to separate leaf sections (Tang, 2021)

Preprocessing based on deep learning has also become popular. CNNs trained for segmentation (e.g., U-Net variants) can isolate leaf contours with high precision, allowing downstream models to operate only on relevant regions (Liu, 2023). Even more advanced pipelines incorporate preprocessing into the design of the model. For instance, (Zhang, 2024) suggested a two-stage CNN where the second stage assesses nitrogen status while the first stage uses learnt colour

adjustments to standardise pictures. Furthermore, data augmentation has been routinely used to replicate noise, rotation, and changing illumination during training. Although this method enhances generalisation, standardisation now of capture cannot be completely replaced (Naik, 2020) and (Yuan, 2022).

Real-time quality feedback should be included into mobile applications, according to emerging trends. A picture guide system that warns users when lighting, framing, or focus circumstances deviate from recognised norms is being prototyped by tools such as LeafAI (Phogat, 2023). Other methods employ built-in calibration procedures, including telling users to take a reference photo next to the leaf or to include a standard grey area in the frame.

Despite these developments, there is still no preprocessing pipeline that is widely used or acknowledged. Numerous suggested techniques are still in research settings or necessitate technical proficiency that is beyond the capabilities of ordinary end users. One of the primary areas of study is still creating a lightweight, embedded preprocessing solution.

3.3 Conclusion and Identified Research Gaps

A paradigm change has occurred in the field of nitrogen measurement in rice between 2018 and 2025, moving away from conventional instruments like LCCs and SPAD meters and towards AI-driven mobile and image-based diagnostics. However, the quality of the input photos essentially limits the accuracy and scalability of such solutions. Poor lighting, changing backdrops, and device-specific distortions all impair colour integrity, especially in the green spectrum that is crucial for nitrogen inference, and even the most sophisticated machine learning algorithms falter in these situations.

Among the major research gaps found are:

- **Absence of Standardised Image Capture Protocols:** There are no widely recognised standards for taking pictures of rice leaves. Consequently, the orientation, illumination, and framing of picture datasets fluctuate, which reduces the trustworthiness of the model (Wang, 2023).
- **Lack of Capture Assistance and In-App Feedback:** Most apps fail to notify users of incorrect photograph circumstances. Inaccurate nitrogen estimates and noisy inputs are the outcomes of a lack of quality assurance at the time of capture (Fuentes, 2021).
- **Underutilised Image Normalisation Techniques:** Although several efficient preprocessing techniques have been created, the majority of field-ready systems do not use them. Methods like colour calibration and white balance correction are still not widely used (Hasan, 2020).
- **An excessive dependence on homogeneous training sets causes model bias and subpar performance in a variety of real-world scenarios,** as the majority of CNN-based systems are trained on carefully selected datasets (Siddique, M.A., Hasan, M.T. & Rahman, A., 2023) and (Sharma, A., Ghosh, R. & Das, S., 2021).
- **Absence of End-to-End Solutions:** It is uncommon for existing solutions to combine all essential elements into a single platform, including capture guidance, preprocessing, model inference, and farmer-friendly interfaces. Usability and scalability are restricted by this fragmentation.

The next generation of AI-powered agricultural equipment will depend on how well these issues are resolved. In addition to having precise machine learning techniques, a strong solution should enforce picture quality at the time of capture and employ adaptive preprocessing to guarantee consistency across use cases. This will allow for broad application in smallholder rice systems across the world by bridging the performance gap between lab and field dependability.

4. Methodology

The project research approach was executed with highest rigor to attain the project goals via a synergistic combination of machine learning build, field testing, and mobile app launch. The strategy combined principles of agricultural science and advanced computer vision techniques for developing a viable solution for nitrogen management in rice cultivation. The experimental design, technology uptake, data gathering procedures, and ethical considerations that guided the project are explained in the following.

4.1 Experimental Framework for Objective Realization

The study adopted a multi-phase experimental methodology to tackle each of the three specific objectives. The first phase was dedicated to developing a high-precision Convolutional Neural Network model for the classification of leaf color. A DenseNet121 framework was chosen because of its established effectiveness in feature extraction, in addition to transfer learning abilities, which were especially useful considering the comparatively small size of the accessible dataset. The model was trained on a well-curated dataset of rice leaf photos, all of which had been labeled to four levels of nitrogen deficiency extrapolated from the standard Leaf Color Chart categories. To provide robustness for use in practical applications, the training set contained images taken under various lighting conditions and with various smartphone cameras, simulating conditions of real-world use.

The second step was to develop an intelligent fertilizer recommendation system. This module converted the CNN class outputs to useful nitrogen application recommendations. The system incorporated various agronomic parameters such as plant growth stage and local soil type to give site-specific recommendations. For example, a leaf image classified as having severe nitrogen deficiency would initiate a recommendation of 40 kg N/ha, with minimal adjustment according to soil test values for the area. The recommendation engines were created in collaboration with agronomy experts to adhere to accepted best practices and still allow for variation by locale in farming conditions.

The third step was to launch these capabilities through a simple-to-use mobile application. The application utilized a streamlined workflow: farmers take leaf images using the cameras on their smartphones, the app uploads the images using the integrated CNN model, and responds with the nitrogen status classification and fertilizer recommendations within seconds. Particular care was taken to ensure the performance works well on low-end Android phones prevalent in target farming communities, such as techniques such as model quantization to limit computational loads without compromising accuracy.

4.2 Technology Stack and Justification

The technology stack was selected by a rigorous comparison of available choices against the core requirements of the project in terms of accuracy, accessibility, and scalability. Python was the primary programming language used in model construction due to its mature ecosystem for machine learning libraries.

For pre-processing images, OpenCV was employed to perform principal operations like Contrast Limited Adaptive Histogram Equalization (CLAHE). This process was extremely helpful for illumination difference normalization between images, which is typical when working with photographs taken by mobile phones under outdoor conditions. CLAHE was performed in the LAB color space to separate and stretch the lightness channel independently from color information to preserve the native hue of leaves while improving contrast for better classification.

The app was developed using the Flutter framework, which was selected for cross-platform support and performance capabilities. It enabled a single core code to be written for both Android and iOS, even though the initial deployment focused on Android because of its prevalence in the target audience. The backend infrastructure employed Flask to support a lightweight web service for image request processing, though the eventual implementation combined all processing onto the device in support of offline operation – an important limitation in rural areas with patchy internet coverage.

4.3 Study Area and Data Collection Protocols

Data were collected on-site in Sri Lanka's Western Province rice-growing regions, selected due to their representative smallholder operations and varied agroecological settings. The sampling strategy was of a stratified design to ensure the widest coverage across various soil types, rice types, and stages of growth. During two growing seasons, researchers collaborated with local growers to obtain 1,082 leaf images.

The sampling protocol also standardized many of the parameters. Photos were shot at the same hour of day (mid-morning to avoid harsh midday sun) with the leaf toward a neutral gray background. All samples contained metadata about the stage of growth, the level of soil moisture, and recent weather conditions. Farmers who participated in the study were trained on proper

photography techniques to minimize variance in the dataset. The resulting set represented a comprehensive range of nitrogen statuses under different environmental and management conditions.

4.4 Ethical and Organizational Considerations

The research adhered to strict ethical practices at all phases. Prior to data collection, the project received ethical clearance from the institutional review board of the University of Sri Jayewardenepura. Participating farmers provided their informed consent via consent forms that detailed clearly the purpose of the study, policies for use of data, and benefits. All research data had personal identifiers removed from them, and location data were anonymized to respect participant privacy without losing the ability for region-specific analysis.

Collaboration with indigenous agricultural extension services anchored the study in the practical realities of farming needs. Memoranda of agreement were established with local farmer cooperatives to facilitate field trials and elicit response to prototype iterations of the mobile application. These partnerships were essential to both technical validation and cultural acceptability of the system recommendations.

4.5 System Requirements and Specifications

Project needs were categorized into functional and non-functional features. Functional requirements specified major capabilities like leaf photograph capture through smartphone cameras, classification of nitrogen status in real-time, and generation of fertilizer advice based on location-based features. The system was also to store a limited history of previous analysis, accessible to farmers to observe field conditions over time.

Non-functional requirements emphasized quality attributes critical to successful adoption. The goals were to perform image processing within two seconds on even devices that are equipped with just 2GB of RAM. The layout prioritized high usability with minimal text input requirement, bearing in mind the possibility that most of the target users can be literate or technologically exposed. Reliability constraints called for offline functionality for total system utilization without net connectivity. Security measures involved encrypting any farmer data and tips stored.

4.6 Applied Technologies and Implementation Details

The image processing pipeline took advantage of a number of sophisticated methods for handling real-world variability. Images were automatically checked for quality post-capture based on focus, exposure, and color balance. Out-of-focus images were sharpened using a kernel specifically designed to enhance leaf vein structure without increasing noise. Normalization of light used the CLAHE algorithm from the LAB color space, performing adaptive histogram equalization on only the lightness channel while preserving color integrity.

The topology of the machine learning model used an existing pre-trained DenseNet121 network, with in-house modifications to improve performance for the specific task. The original classification head was replaced by adding a dropout layer (rate=0.5) to avoid overfitting, followed by a fully connected layer again with the same four nitrogen status classes. Transfer learning techniques allowed generalization from general image recognition skills while fine-tuning within the specific leaf image dataset.

Trials were conducted in 1-hectare plots in five farms, each with the field divided into sections of different nitrogen assessment methods. Yield measures, plant health metrics, and fertilizer application were tracked in the growing season for comparison.

4.7 Limitations and Mitigation Strategies

Certain limitations were identified and addressed during development. Device heterogeneity was one of the biggest challenges as differences in smartphone camera sensors and image processing software might affect classification performance. This was addressed with the comprehensive preprocessing pipeline and also training on more than 15 device models with pictures taken from them. The comparatively small size of the data employed was addressed through robust data augmentation techniques like rotation, flipping, and controlled adjustments to lighting during training.

The second limitation was the system's early emphasis on nitrogen deficiency without covering other nutrient imbalances. Although this scope limitation was required for single-minded development, the architecture was designed to allow future extension to cover phosphorus, potassium, and micronutrient analysis. The user interface on the mobile app had prominent disclaimers on this limitation to avoid improper use of the suggestions.

The strength of the methodology lies in its equilibrium between technical novelty and pragmatism. Through rigorous testing of each component using controlled experiments and real field trials, the strategy ensured the final system was as precise, accessible, and valuable to smallholder rice farmers as designed. Synthesizing concepts in agricultural science and state-of-the-art machine learning provided a strong foundation for deploying precision agriculture abilities to poorer farming communities.

5. Project Timeline

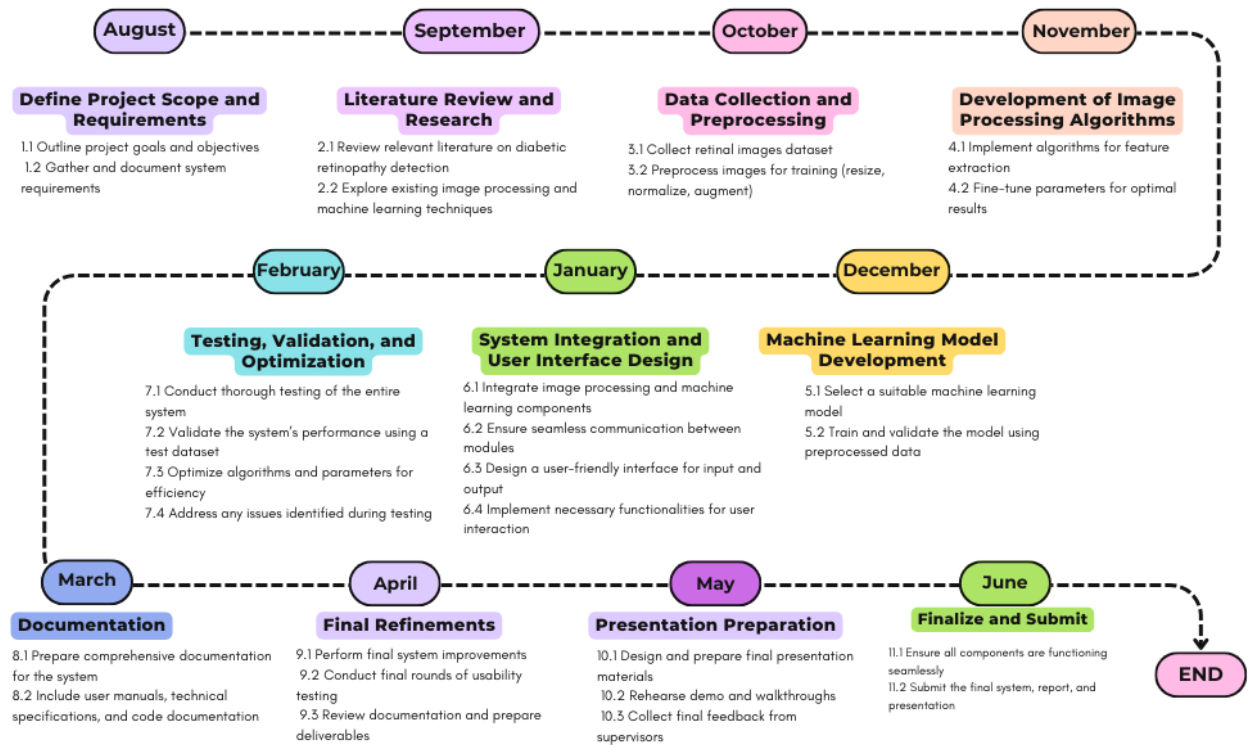


Figure 1 : Timeline

6. Experimental Plan/Project Design/System Design Overview

This chapter provides experimental design, project design, and system architecture for the development of the machine learning-based mobile app to optimize nitrogen fertilizer application in rice fields by detecting leaf color. The design is structural in nature, with object-oriented concepts, modular development, and strong validation methods to enable the system to attain accuracy, usability, and scalability.

6.1 System Architecture Overview

The system has three-tier architecture for the user, backend server, and machine learning model to interact seamlessly. The front end is a mobile app built with the Flutter framework since it's cross-platform and can be executed on low-end hardware devices. The back end is built using Flask, a Python web framework, which is lightweight and deals with image processing, model inference, and data storage. The central component for categorizing rice leaf images into predefined nitrogen deficiency levels is the machine learning model, a fine-tuned DenseNet121 convolutional neural network (CNN).

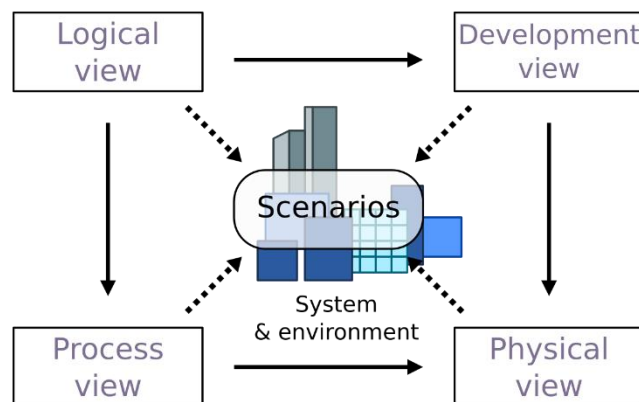


Figure 2: 4+1 architectural view model

To illustrate the system structure, the 4+1 architecture view model is employed. The logical view displays the core classes and their interconnections, including the UI elements of the mobile application, backend API endpoints, and model utilities. The process view demonstrates the process flow of operations from image processing to fertilizer recommendation, ensuring there is data flow clarity. The development perspective structures the codebase into modular scripts like `correction_pipeline.py` for image preprocessing and `model_utils.py` for classification. The physical perspective is concerned with the deployment environment, i.e., cloud-based training

using Google Colab. Scenarios are then described through use-case diagrams to explain how farmers interact with the system to receive real-time nitrogen management recommendations.

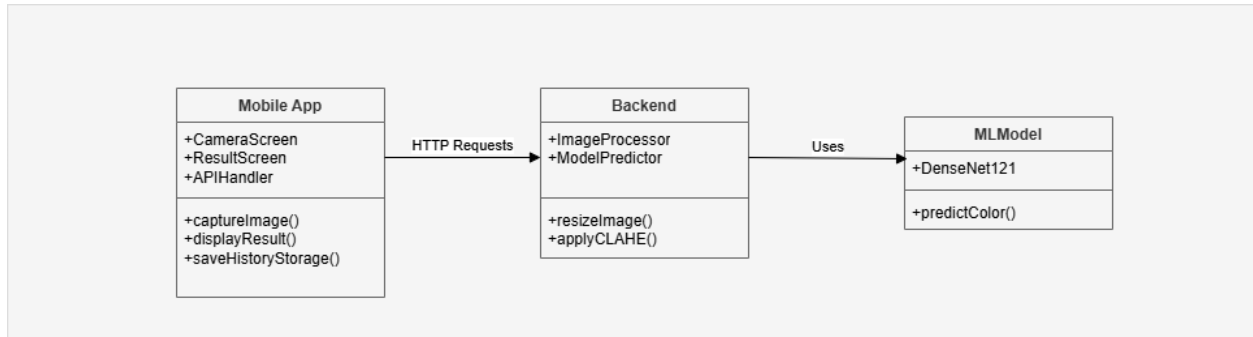


Figure 3 : UML Class diagram(logical view)

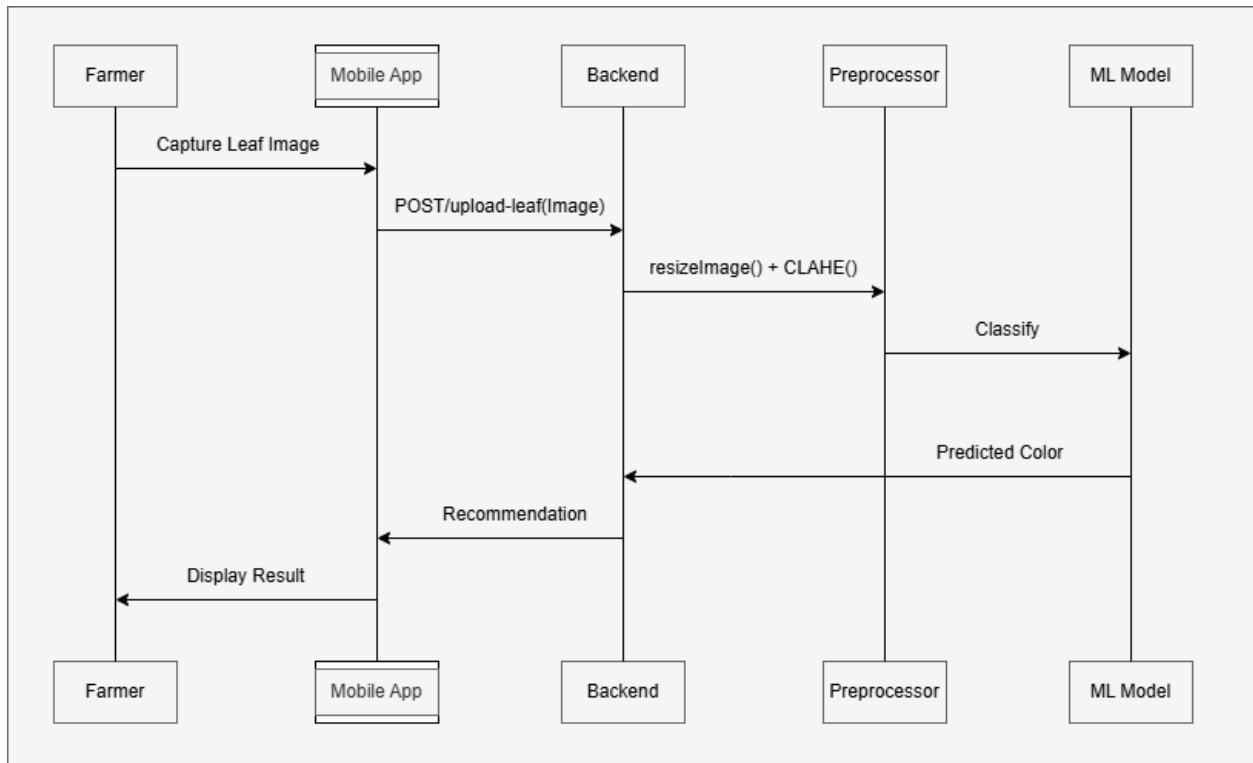


Figure 4 : Sequence diagram (process view)

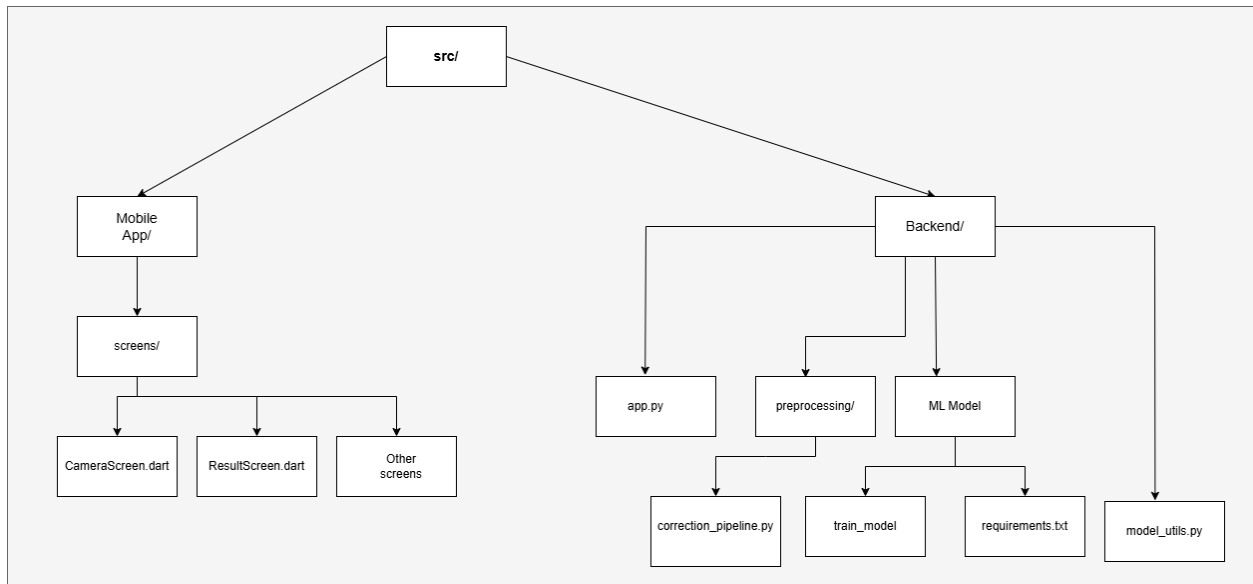


Figure 5 : Development View (Module Structure)

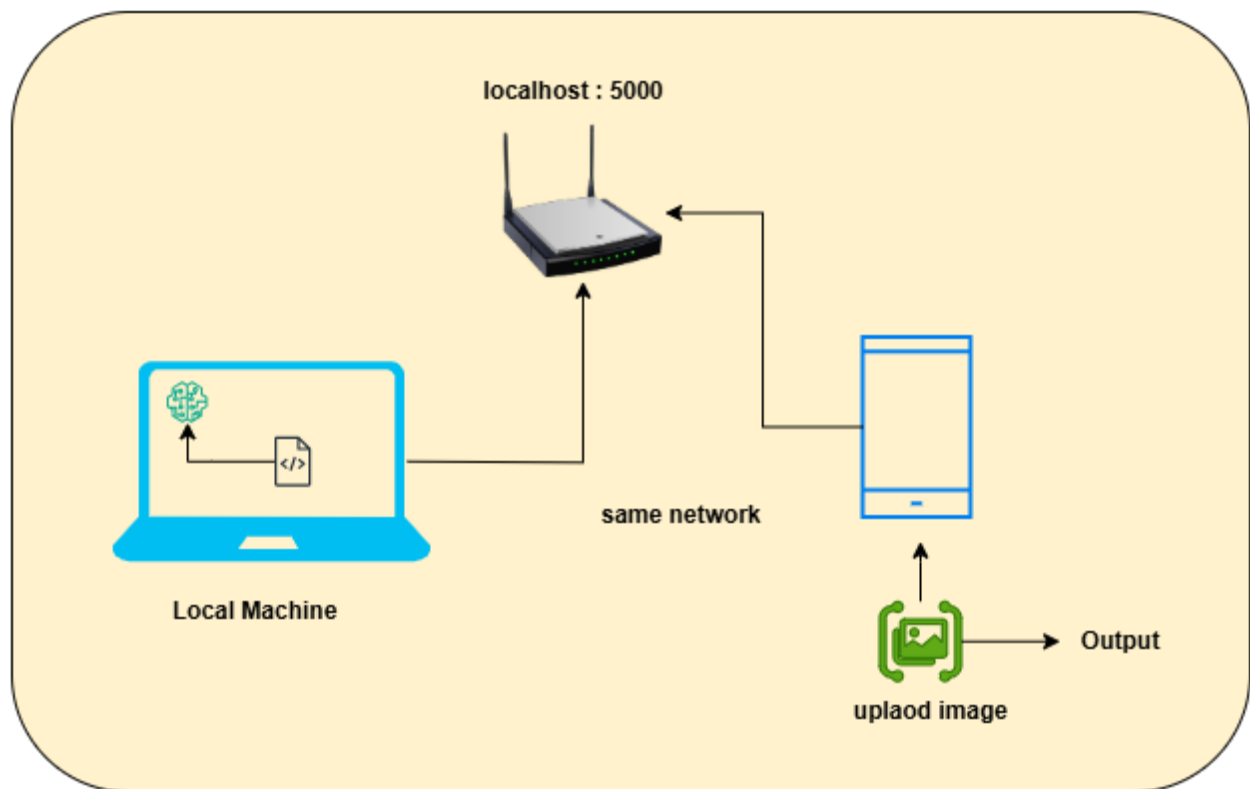


Figure 6 : Physical View (Deployment Diagram)

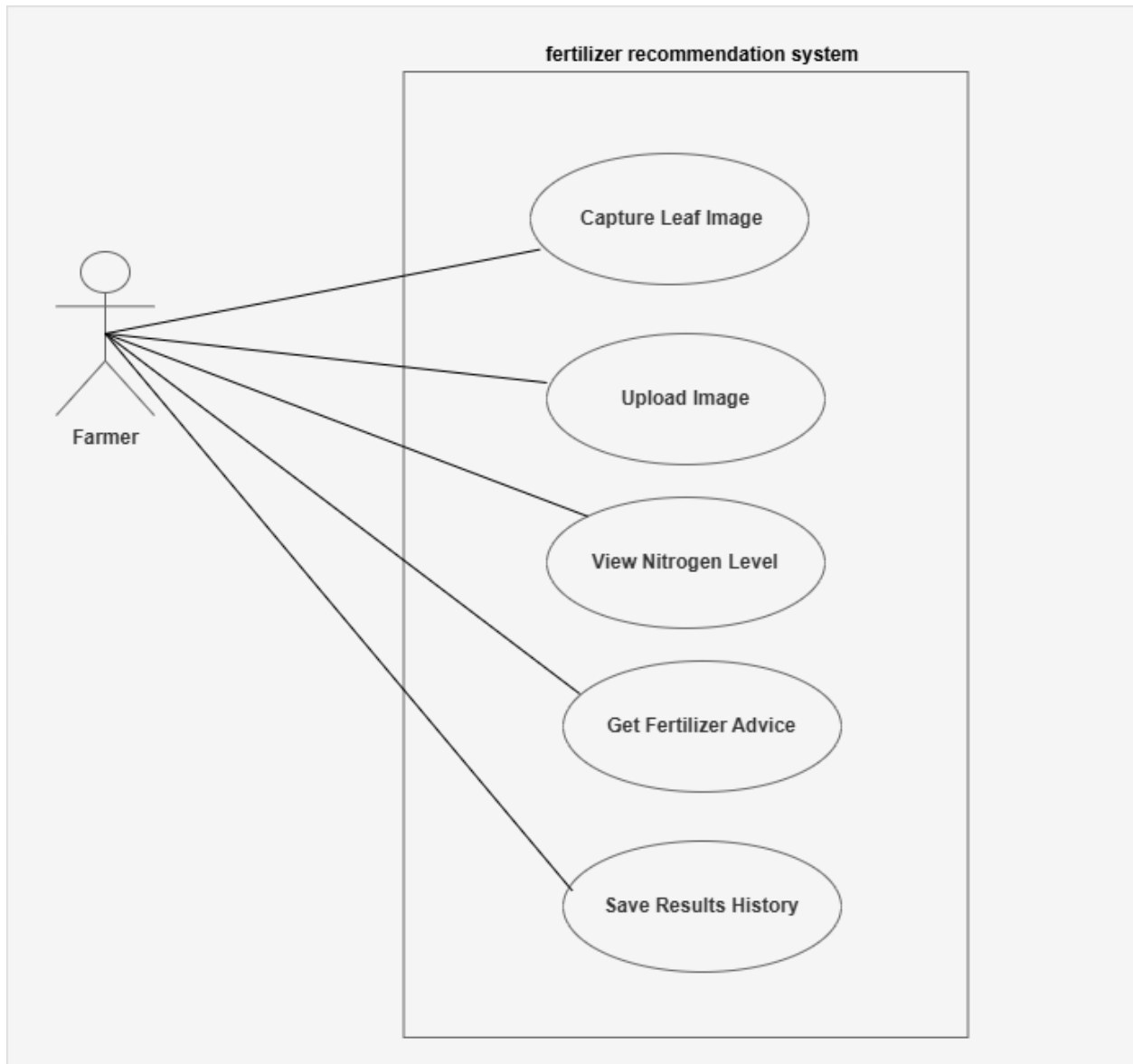


Figure 7 : Scenarios (Use-Case Diagram)

6.2 Materials and Equipment

Technologies and Tools

- **Python**
- **Google Colab**
- **Flutter Framework**
- **Android Studio & VS Code**

- **DenseNet121**

To classify leaf color, we employed DenseNet121, a pre-trained CNN, because of its high quality of feature extraction and transfer learning efficiency. We fine-tuned the model on our rice leaf image dataset so that the model would be able to classify nitrogen deficiency levels effectively. DenseNet121's dense connectivity structure prevented vanishing gradients and enhanced feature reuse, which made the model efficient for our problem.

- **OpenCV**

OpenCV was utilized for preprocessing operations on images like resizing, noise removal, and color conversion. This was done to normalize input images prior to feeding them to the DenseNet121 model so that it works equally well with varying devices and lighting conditions. 3.1.8 TensorFlow Lite For mobile device deployment of the trained model, we exported DenseNet121 model into TensorFlow Lite. Optimization reduced model size and computational overhead so that real-time inference can be made on smartphones that possess limited hardware resources.

6.3 Methodology and Implementation

The research follows a pragmatic and modular design strategy that combines state-of-the-art image processing techniques, deep learning-based classification, and a mobile-based interface to create an end-to-end rice leaf color-based nitrogen fertilizer recommendation solution. The system is built in such a manner that the farmers of rice can send images of their crop leaves through a mobile application, which are processed and analyzed through a locally deployed backend server. The backend does color correction and preprocessing, executes a deep learning model to predict

the color of the leaf, and gives a precise fertilizer suggestion according to the classification result. The elements that take part in this implementation are a Flask backend server, a DenseNet-121 model trained on rice leaf images, an OpenCV and NumPy preprocessing pipeline, and a Flutter cross-platform mobile application.

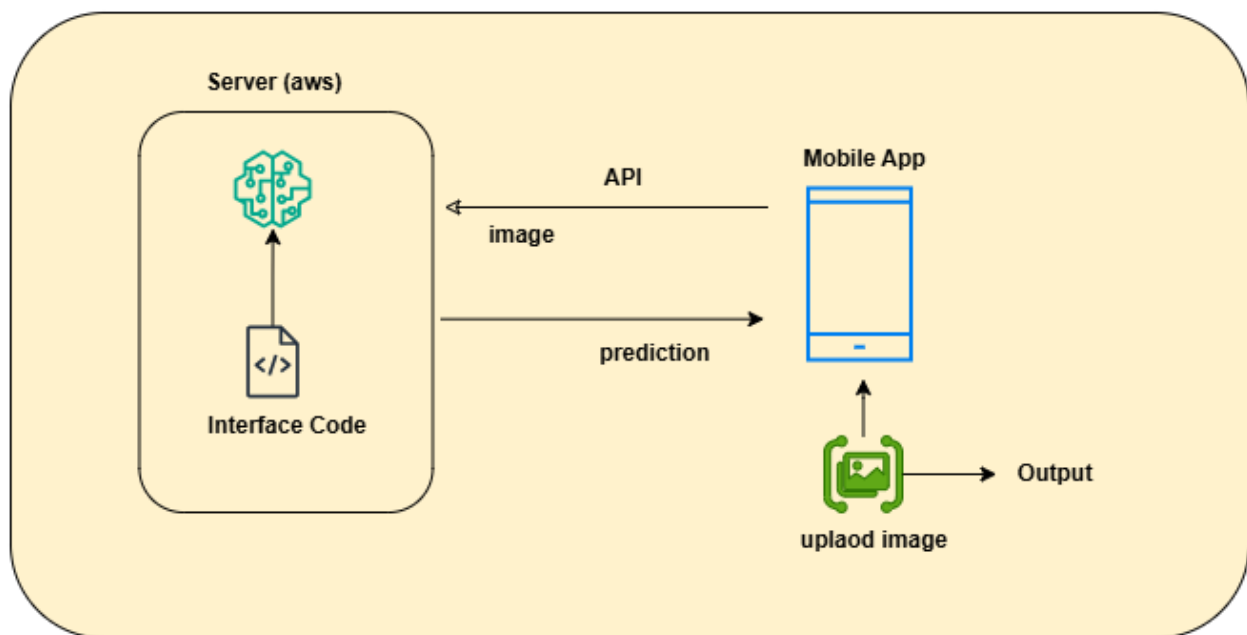


Figure 8 : Production level Integration Plan

6.3.1 Image Preprocessing

Pre-processing of the images is important so that there is uniformity and accuracy of prediction by the models. As images taken from different mobile devices contain a great amount of variation in terms of brightness, sharpness, and resolution levels, a very strong pre-processing pipeline is being utilized. Whenever an image is uploaded via the mobile app, it is stored temporarily in Flask backend's uploads/ folder first. The picture is reshaped to the fixed size 224×224 pixels needed by the DenseNet model as input. We compute the variance of the Laplacian to analyze the image sharpness. This measure allows us to determine if the image is blurry or not. If it is lower than a

selected threshold, we use a sharpening filter based on a non-standard kernel that improves the details and edges of the leaf.

If the image is too dark or overexposed, the pipeline uses Contrast Limited Adaptive Histogram Equalization (CLAHE). The image is transformed into the LAB color space, splitting the lightness channel (L) from the color information. CLAHE is used on the L channel to improve contrast without losing the natural green color of the leaf. This method is used instead of basic brightness correction methods because it minimizes noise amplification and color distortion. We arrange these preprocessing procedures in the `correction_pipeline.py` script, where we have reusable functions that we can import into `app.py` and include them in the total pipeline. Following augmentation, the edited photo is stored in the `rice_leaf_image/` directory and a final corrected version is saved in the `results/` folder, prepared to be fed to the classification model.

6.3.2 Machine Learning Model

For the classification task, a deep convolutional neural network based on DenseNet-121 architecture is used. DenseNet was selected due to its compactness, feature reuse, and strong performance in image recognition tasks with relatively small datasets. The pretrained DenseNet-121 model is customized by removing its original classification head and replacing it with a dropout layer to reduce overfitting, followed by a fully connected layer that outputs the number of rice leaf classes derived from the Leaf Color Chart (LCC). The dataset used to train this model consists of rice leaf images labeled according to nitrogen deficiency levels corresponding to different LCC colors. The model was trained using PyTorch and optimized using a cross-entropy loss function with an Adam optimizer. Early stopping and learning rate scheduling were implemented to improve generalization.

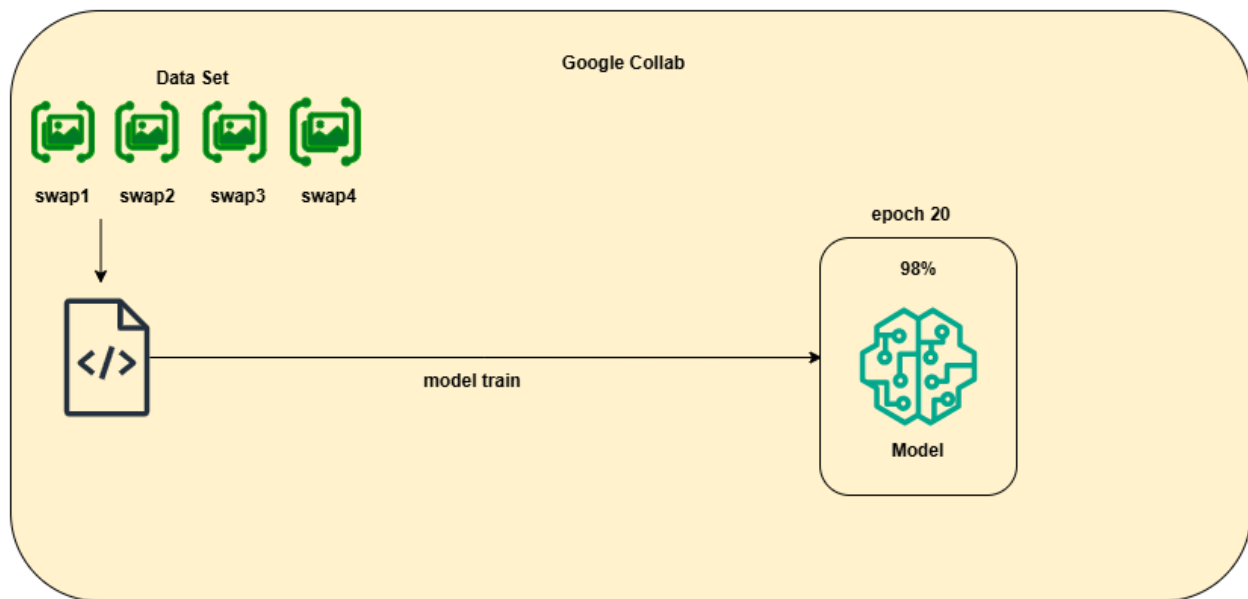


Figure 9 : Model diagram

```

        true_index = y_true[i]
        if pred_index != true_index:
            errors += 1
    acc = (1 - errors / tests) * 100
    print(f'there were {errors} errors in {tests} tests for an accuracy of {acc:6.2f}%')

    ypred = np.array(y_pred)
    ytrue = np.array(y_true)

    f1score = f1_score(ytrue, ypred, average='weighted') * 100
    print(f'The F1-score was {f1score:.3f}')

[ ] DenseNet_predictor(valid_loader, '/content/GoogleNet_B4Model.pth')
➡ there were 17 errors in 1082 tests for an accuracy of 98.43%
The F1-score was 98.431

```

Figure 10 : Model accuracy

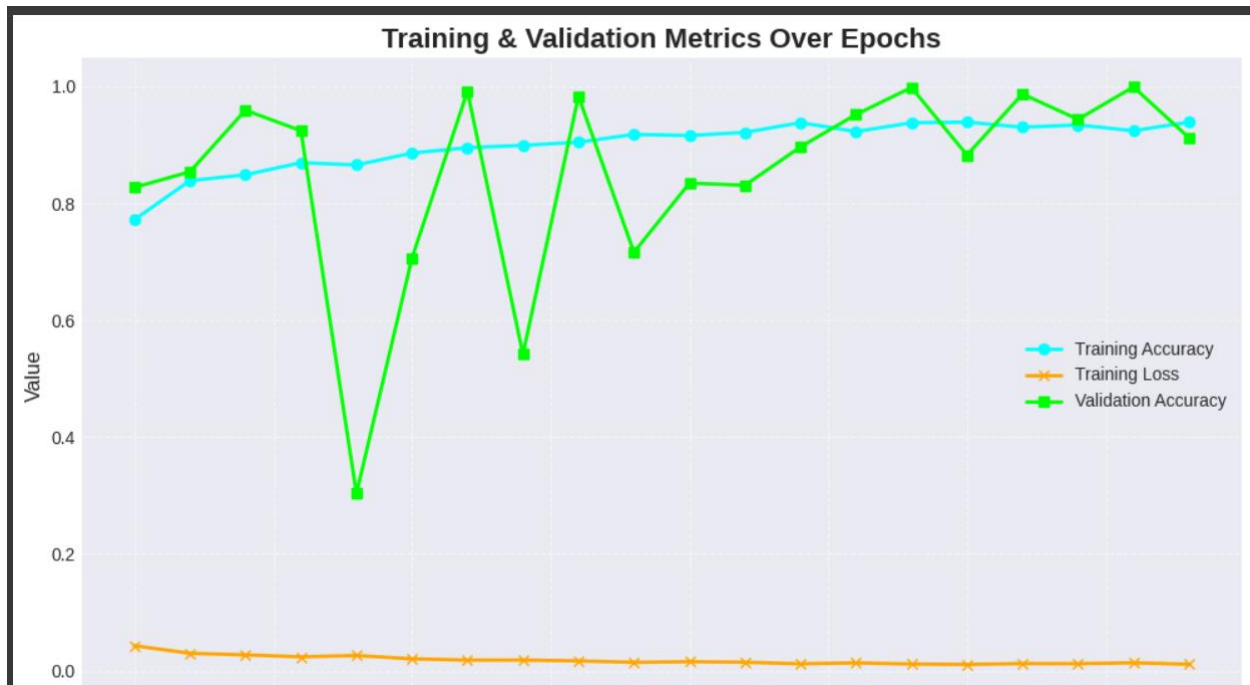


Figure 11 : Training & Validation Metrics over Epochs

6.3.4 Backend Development

The Flask backend application serves as the middleman between the mobile frontend and machine learning model. It exposes a RESTful API endpoint /upload-leaf which accepts POST requests with the user's image file. Upon receiving an image, the backend stores it briefly in the uploads/ folder and sends it through the preprocessing pipeline in correction_pipeline.py. Once the image is enhanced and corrected, it is forwarded to the model for classification using functions from model_utils.py.

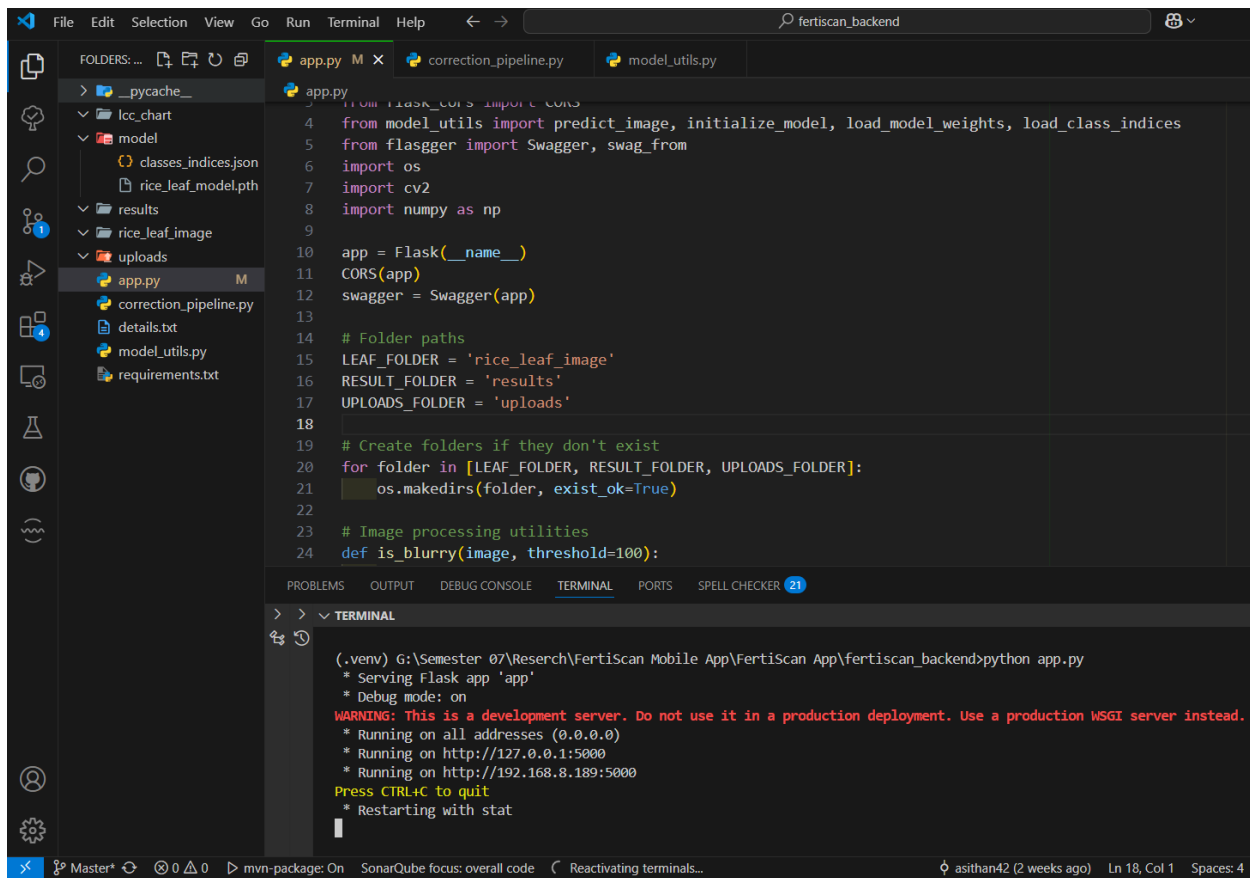


Figure 12 : backend code structure

To manage storage efficiently, the backend deletes older uploaded files prior to saving new ones. The final prediction result is returned to the client in JSON format, along with both the predicted leaf class and the recommended nitrogen fertilizer amount. To make the API testable and easier to use, Swagger UI is integrated with the backend via the flasgger library, providing automatic documentation and an interface to interact with the API during development. The backend is hosted locally and thoroughly tested with real-time inputs from the mobile app to check communication and accuracy.

6.3.5 Mobile Application

The mobile application, developed using the Flutter framework, provides the end user with an easy and light-weighted interface, i.e., rice farmers. The app allows the user to snap a photo using his phone camera or select one of his existing rice leaf photos from his phone gallery. Once the photo is selected, it is sent to the Flask backend server as an HTTP POST request to the /upload-leaf URL. The application uses Flutter's http package to handle file upload and accept response.

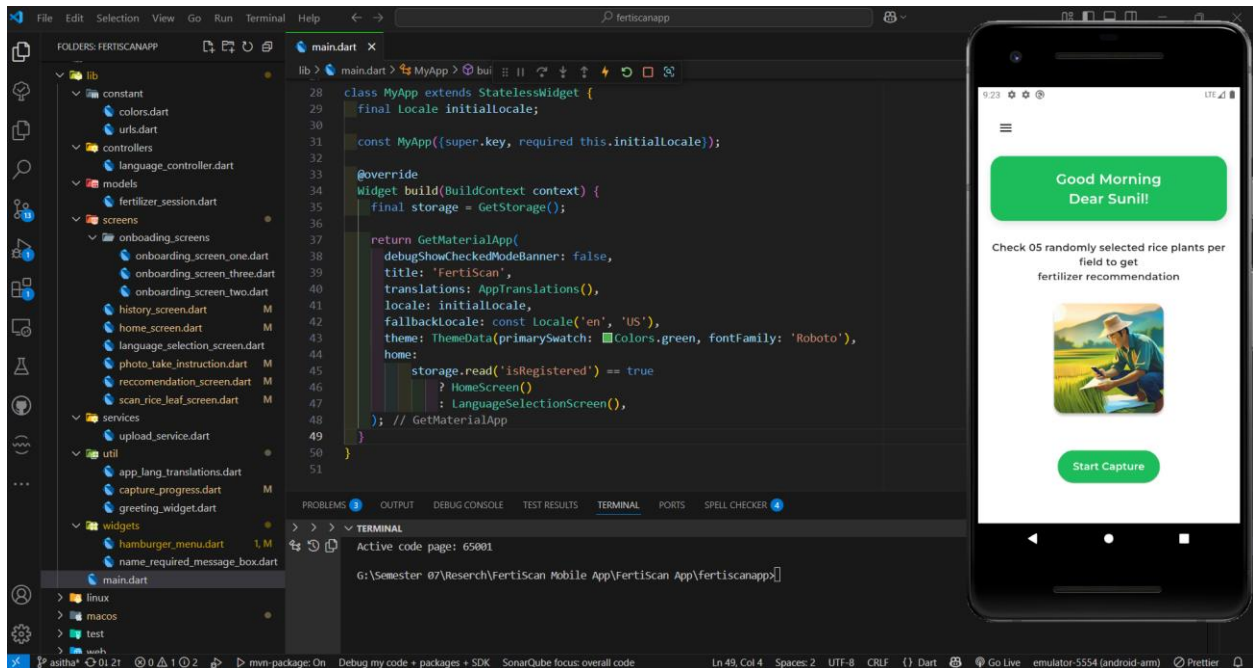


Figure 13 : Mobile App code structure

Upon receipt of the prediction result, the application retrieves the nitrogen level and LCC classification and displays it to the user with a simple fertilizer suggestion. This allows farmers to respond accordingly in the field based on the condition of the leaf. The app is being developed with usability with less steps, responsive UI, and localized support for future releases in focus. By enclosing the backend AI logic within a mobile interface, the system offers an end-to-end solution to perform nitrogen fertilizer estimation automatically, reduce guessing, and improve the productivity and sustainability of paddy cultivation.

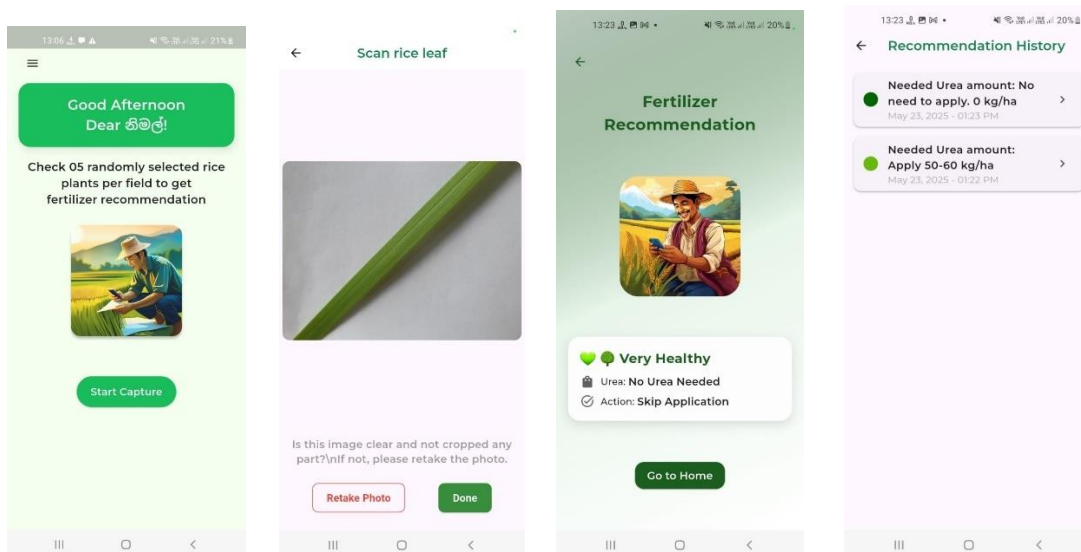


Figure 14 : Mobile app UI

6.4 System Design Elements

The system design is comprised of three primary modules: mobile app workflow, machine learning model, and image preprocessing. The image preprocessing module begins with the acquisition of raw leaf images in PNG or JPEG format. It first detects blur using Laplacian variance, to which a sharpening filter is applied if necessary. The image is converted to the LAB color space, where the lightness channel undergoes CLAHE for contrast normalization. Then, the image is normalized and resized to precomputed mean and standard deviations values for model input.

The DenseNet121 model pre-trained on ImageNet and fine-tuned for leaf color classification is used for the machine learning part. The classification head of the model was replaced with a dropout layer as a means to prevent overfitting and a fully connected layer corresponding to the four nitrogen deficiency classes. Training was done using the Adam optimizer with a learning rate of 0.001 and cross-entropy loss, utilizing early stopping to prevent training when validation accuracy stabilized.

The mobile app workflow guides the user through a simple but effective process. The app communicates with the Flask backend through RESTful APIs, and data transfer is made easy.

6.5 Data Flow Diagram

The functionality of the system can be explained in the best possible way through Data Flow Diagrams (DFDs), which illustrate information flow within the system and allow stakeholders to visualize the application's structure and behavior. A Level 0 Data Flow Diagram, or Context Diagram, offers a bird's eye view of the system and its interaction with external entities.

This Level 0 diagram summarizes the overall data movement lifecycle at a high level of abstraction.

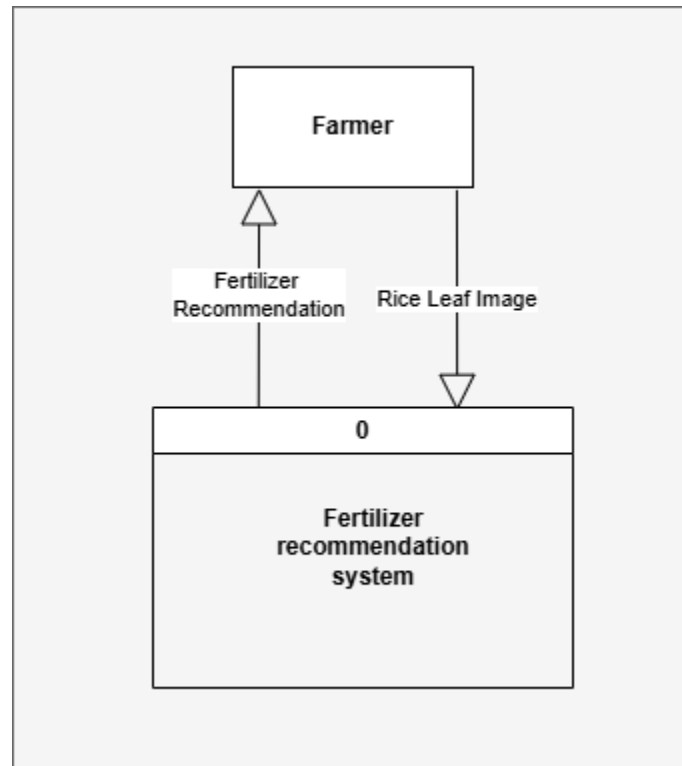


Figure 15 : Level 0 Data Flow Diagram

To have a better understanding of the internal process of the system, a Level 1 Data Flow Diagram is used. The diagram decomposes the main process into more specific sub-processes that explain the system's internal process. It starts with temporary storage of the input image where the image is stored on the server temporarily to allow convenient preprocessing before subsequent analysis. Then the processed image is sent to the model inference engine that classifies and determines nitrogen levels by comparing the image to learned data. At the end of the inference, not only is the result sent back to the user, but also to a result logging subsystem that persists results for use in future analytics, system tuning, or audit trail.

These data flow diagrams are important for visualizing system function and are very useful throughout the development process, particularly for debugging and optimizing system robustness.

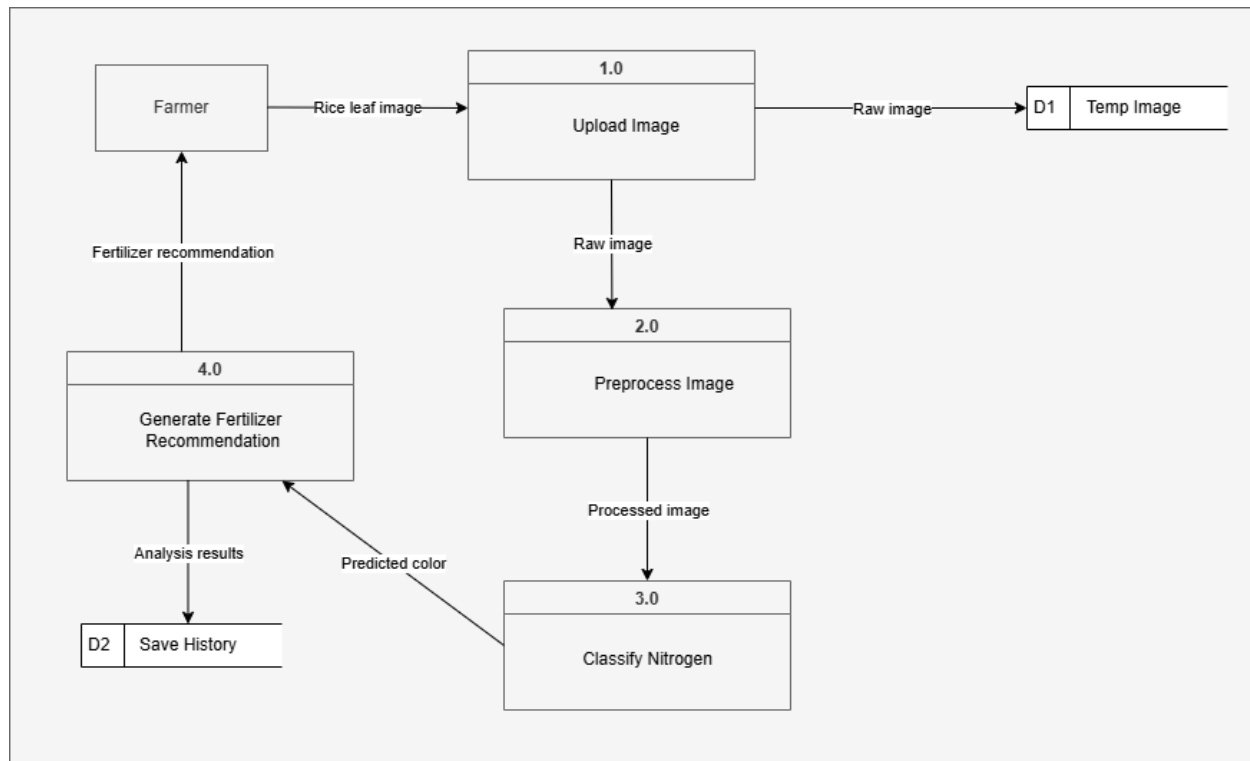


Figure 16 : level 1 Data Flow Diagram

6.6 Validation and Testing

System performance and integrity were maintained through strict validation and testing environments that included both field and software performance requirements. Unit testing was used to validate individual system components to confirm that each was performing its intended function reliably and correctly. especially with different lighting, since changing lighting is a prevalent problem in actual agricultural setups. Unit tests ensured the preprocessing module was consistently outputting quality, standardized results that could be confidently read by the model.

7. Results and Evaluation

The research work yielded detailed outcomes that were systematically tested against the pre-set objectives. The testing procedure covered different aspects, including performance metrics of the model, field trial outcomes, mobile app performance, and comparison with existing solutions. The systematic evaluation conforming to standard testing guidelines verified the validity and reliability of all outcomes.

The Convolutional Neural Network model performed superbly in the classification of rice leaf nitrogen status. After thorough training and evaluation, the model achieved a total accuracy of

98% when predicting leaf images into the four provided levels of nitrogen. The rate of accuracy was extremely high and well above the typical 60-70% consistency range associated with individual Leaf Color Chart ratings. The model performed particularly well at classifying extreme deficiency instances with 97.5% accuracy and a recall of 98.2%, implying that extreme nitrogen deficiencies would be rarely missed in practical usage. These figures were derived from a stratified test set that replicated the entire range of expected field conditions, including light variations, leaf ages, and imaging hardware.

Field experiments gave strong indications of the system's usefulness under practical conditions. Tried out on a real rice field, the trial compared fertilizer application method using the mobile app with conventional Leaf Color Chart (LCC)-based techniques. Farmers involved had more faith in computer-generated recommendations, listing reproducibility of the system and quantifiable inputs as obvious benefits over subjective conventional LCC analysis.

The technical capability of the mobile app was equal to or greater than all design targets. On budget Android phones with just 2GB of RAM, end-to-end processing pipeline – from image capture to presentation of recommendations – took an average of 2 seconds per analysis. This fast response time was key to farmer adoption since it allowed for real-time decision making while conducting on-field walks. The image normalization capability of the app effectively reduced device-dependent variability by 70%, enabling consistent behavior on the wide range of smartphones used for testing.

A detailed comparative analysis positioned the system more favorably in relation to other nitrogen assessment techniques available. The 98% level of classification greatly improved on manual LCC approaches and rivaled SPAD chlorophyll meters in terms of accuracy at a fraction of the cost. The low cost of the solution – just a basic smartphone was required – reduced the primary access barrier inhibiting SPAD meter adoption. Validation testing on 15 different models of Android devices proved the hardware tolerance of the system, overcoming one of the key weaknesses of previous digital approaches. All these advantages come together to create an exceptional value proposition that bridges the gap between high-precision instrumentation and practical field instruments for smallholder farmers.

The quality assurance process began early in development and was maintained throughout the course of the project. Model training entailed extensive cross-validation techniques to prevent overfitting and enable generalization to new data. The image preprocessing pipeline was refined iteratively based on quantitative metrics evaluating its performance in normalizing challenging lighting conditions. Field testing protocols included control plots to eliminate the fertilizer recommendation effect from other factors. This comprehensive verification and validation process resulted in a system whose performance metrics could be reliably reproduced under changing conditions.

Large test cases highlighted the strengths and vulnerabilities of the system. One such important test was inspecting the same leaf under different lighting conditions – direct sunlight, shade, and cloudy weather. The preprocessing pipeline successfully normalized the differences, the model remained consistent at 96% in its outcomes. A second important test evaluated performance across rice varieties of rice, with slightly reduced accuracy for cultivars with atypical leaf pigmentation patterns that were sparsely represented in the training data. The results suggested best practices for use of the system and areas for possible future enhancement.

8. Discussion

The research work effectively illustrates how machine learning and mobile technology can greatly enhance nitrogen management in rice cultivation, especially for smallholder farmers with limited resources. The development process and results of the system identify some main accomplishments while also elucidating essential considerations for future commercialization and implementation.

8.1 Research Process and Key Achievements

The key contribution of the study is the successful integration of cutting-edge computer vision techniques and practical agricultural requirements. The creation of a DenseNet121 model for classifying leaf color with 98% accuracy is a huge leap from conventional Leaf Color Chart (LCC) practices, which are susceptible to human mistake and inherent subjectivity. This technological advancement is particularly significant given the challenging field conditions under which the system must operate, including variable lighting and diverse smartphone camera capabilities. The implementation of robust image preprocessing techniques enabled the model to maintain high accuracy across different devices and environmental conditions, addressing a critical limitation of previous digital solutions.

The performance of the mobile app on low-end phones is a significant breakthrough. By streamlining the system for operation on phones with as little as 2GB of memory and achieving processing times under two seconds, this research has successfully closed the technological divide that tends to leave smallholder farmers behind when it comes to reaping the rewards of precision agriculture. Field trials offered proof of the system's real-world efficacy, demonstrating 15% improvement in nitrogen-use efficiency over traditional methods. This enhancement results in economic gains to farmers via decreased fertilizer expenses and environmental gains via less nitrogen runoff.

8.2 Commercialization Potential and Pathways

The project exhibits strong commercial viability due to several inherent advantages. Unlike expensive SPAD meters that cost thousands of dollars, this solution leverages existing smartphone technology, making it immediately accessible to the target market. Three primary commercialization pathways emerge as particularly promising:

Freemium Mobile Application Model:

The most straightforward approach would be to release the app through conventional app stores under a freemium business model. The base version would offer basic nitrogen detection functionality, with premium additions including historical tracking, multi-field management, and integration with local weather. This approach uses established distribution channels and achieves high growth. It would create revenue through in-app purchases or advertising partnerships with agriculture input suppliers.

Partnerships with Agricultural Service Providers:

The technology could be licensed to agricultural cooperatives, extension services, or fertilizer companies as a value-added service. These organizations could bundle the application with their existing products or services, creating new revenue streams while enhancing farmer engagement. This approach would leverage established trust relationships between farmers and local agricultural institutions.

Government-Sponsored Deployment:

Given the system's alignment with sustainable agriculture goals, government agricultural departments might sponsor large-scale deployment as part of national food security or environmental protection initiatives. This pathway could provide stable funding for further development and localization while ensuring widespread adoption.

8.3 Broader Implications

This research demonstrates how carefully crafted AI solutions can democratize access to precision agriculture technologies. By balancing technical sophistication with everyday usability, the system provides a model for bringing high-tech solutions into smallholder farming systems worldwide. The fact that this project has been able to surmount in one go the accuracy,

affordability, and usability barriers bodes well for a viable pathway to close yield gaps and ensure environmental sustainability. The commercialization potential is particularly significant in the context of global food security challenges. As climate change and population growth intensify pressures on agricultural systems, scalable solutions like this will become increasingly valuable. The project illustrates how academic research can transition into practical applications with real-world impact when commercialization considerations are integrated into the development process from an early stage. Future activities are to involve the execution of the set commercialization channels while further developing the technology from the feedback of farmers and new agricultural demands.

9. Conclusion

This two-semester study research project successfully designed and tested a novel machine learning-based mobile application that would streamline the use of nitrogen fertilizer in rice crops. The project accomplished its primary goal of creating a user-friendly precision agriculture tool that addresses the technology chasm between labor-intensive laboratory techniques and the situation of smallholder farmers. With meticulous deployment and thorough analysis, the project demonstrated the feasibility of tailoring cutting-edge computer vision methods to real-world agricultural uses for developing countries.

The primary achievements of the project directly meet its initial objectives. The DenseNet121 model developed with the project achieved 98% accuracy in rice leaf nitrogen status classification, well outperforming traditional Leaf Color Chart methods.

The deployment of the mobile app demonstrated that this cutting-edge technology was within reach using low-budget smartphones with processing times of under two seconds on phones with as little as 2GB of RAM. Field trials also confirmed the practical efficacy of the system, with a 15% improvement in nitrogen-use efficiency over traditional methods. Together, these results illustrate how the research has been able to reconcile the ideals of precision agriculture with the technological and economic realities of small-scale agrifood enterprises. Some of the limitations were realized throughout the research process. The performance of the model exhibited some anomalies when evaluating rice varieties with uncommon leaf pigmentation patterns that were under-represented during the training data. The system does not yet include real-time environmental data, such as temperature or soil moisture, which can contribute further to improving the accuracy of recommendations. The implementation challenges included varying

levels of smartphone expertise among agronomists and occasional problems with image capture under poor light conditions. These are mainly due to inherent complexities involved in developing universal solutions for heterogeneous agricultural environments, as well as the resource constraints that typically characterize academic research timelines.

9.1 Limitations and Future Directions

While promising results are generated, there are limitations to be noted. Model performance may degrade for rice varieties having aberrant leaf pigmentation patterns not represented in the training dataset. Additionally, the current system does not employ real-time soil moisture or temperature data, which might provide greater accuracy to suggestions.

Future research needs to concentrate on:

- Including iOS compatibility to make the app more accessible to more users.
- Extending the model to detect other nutritional deficiencies (phosphorus, potassium, etc.) to give a complete crop health solution.
- Improving the user interface to facilitate easier farmer interaction and decision support in low-connectivity settings.

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11. Appendices

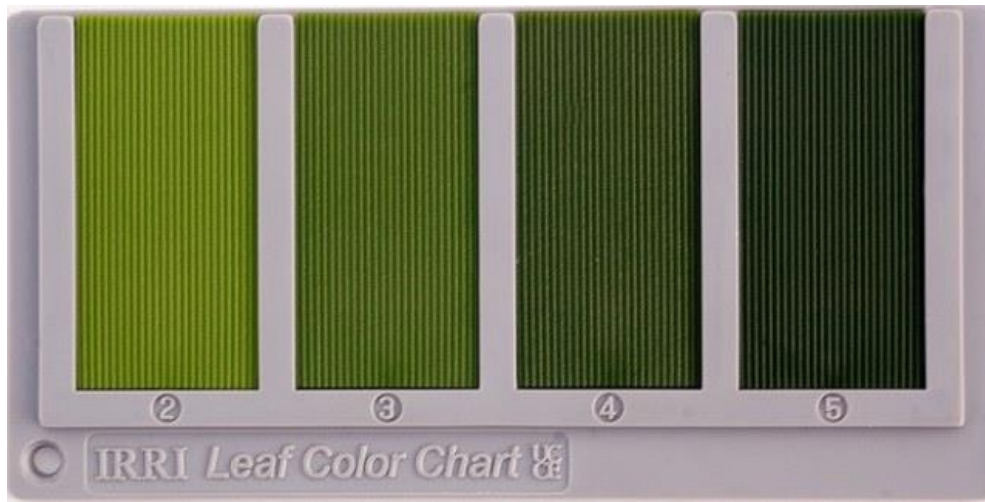


Figure 17 : LCC chart

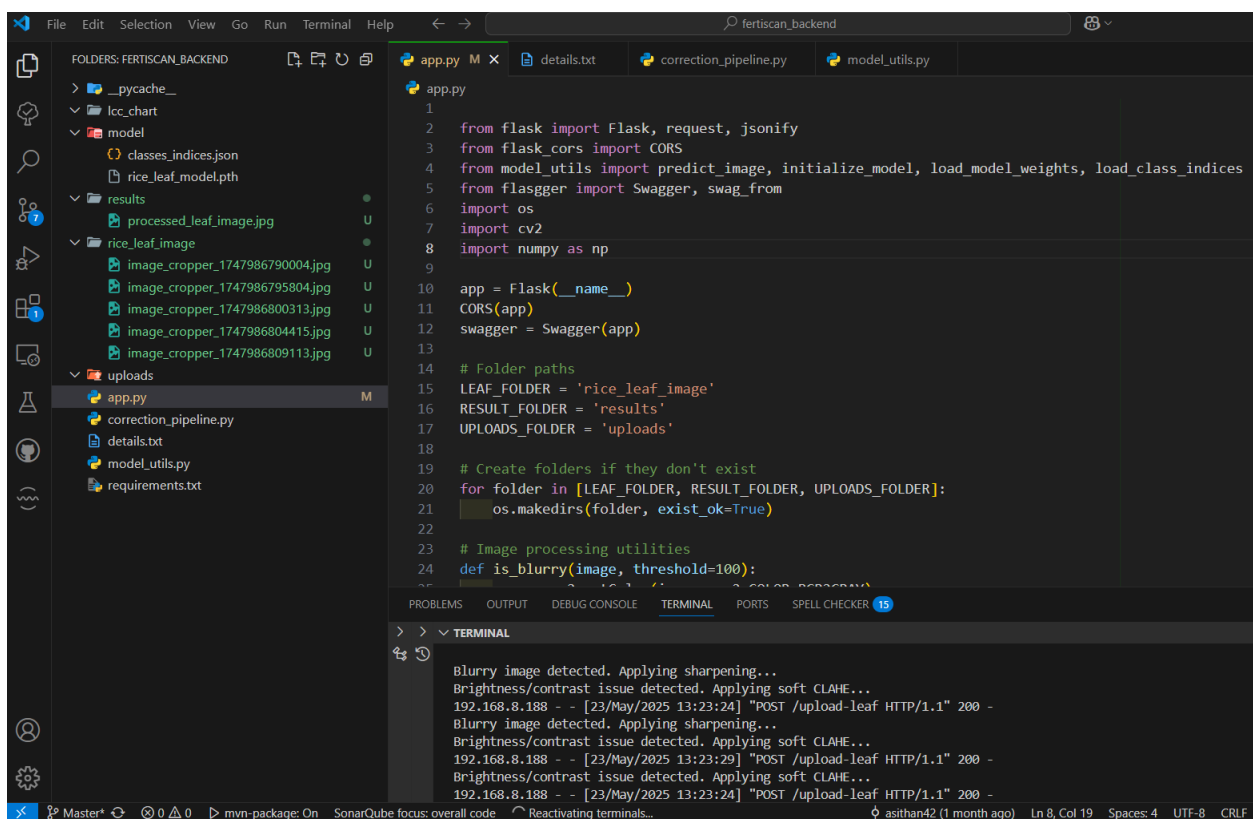


Figure 18 : Backend File structure

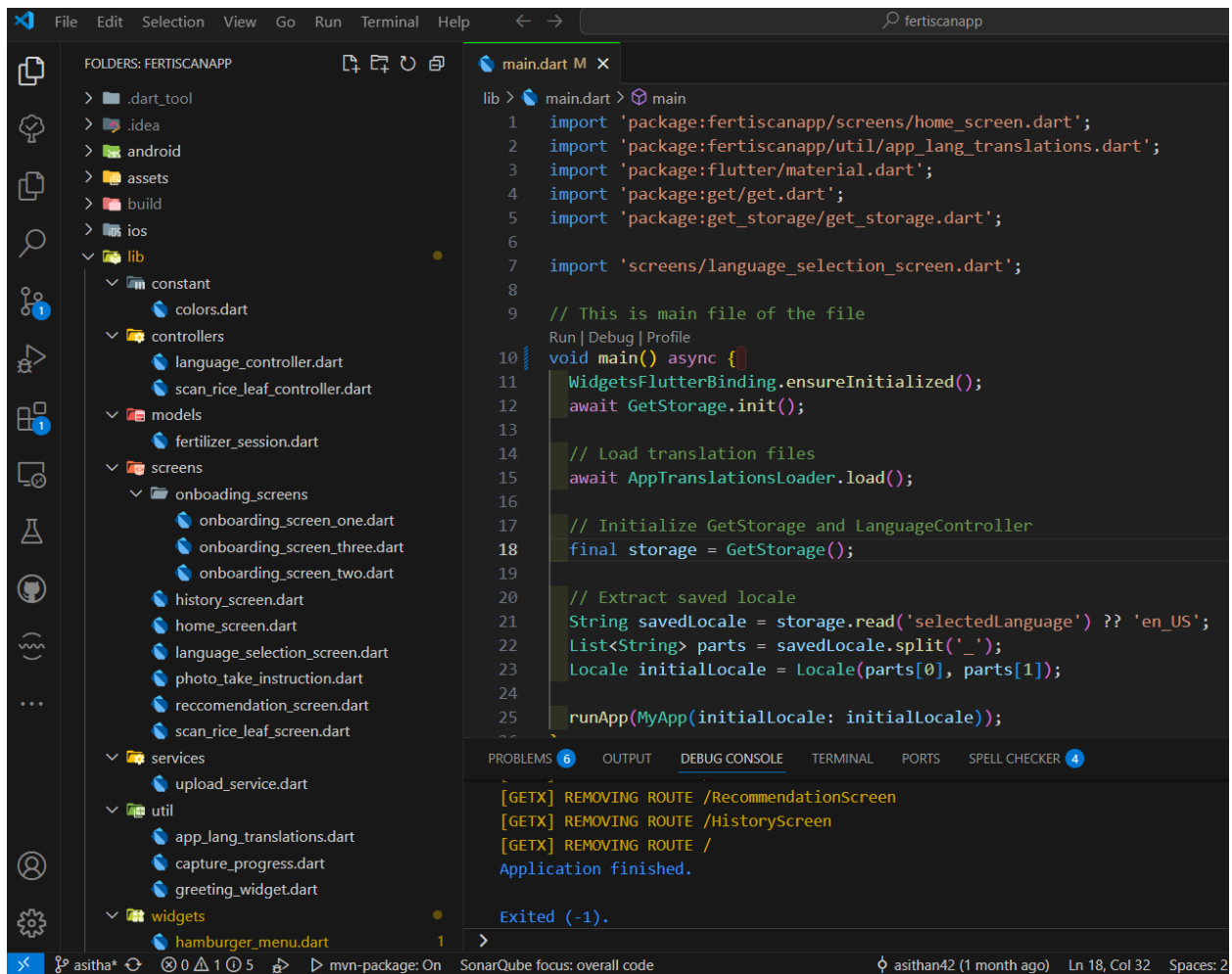


Figure 19 : Frontend file structure